

fore, it is not evident why larger or more manholes are necessary for this purpose on the translator system.

The translator system may be used with single-conductor cable with two of the phase wires larger than the third—just as is done in radial practise. These sizes would not have to be changed from block to block but would be uniform over the system. The case of power load equaling half the lighting load required that the wire carrying the maximum current in the translator system use about 34 per cent more copper than for the star system instead of 60 per cent as given by Mr. Richter.

Mr. Kehoe refers to a scheme using transformer secondaries. Such a connection is shown in Fig. 1, herewith, utilizing four

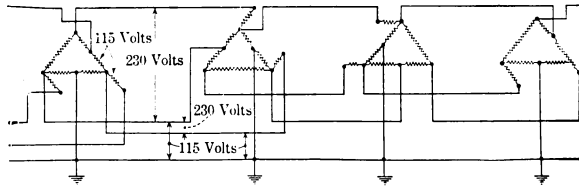


FIG. 1

standard distribution transformers. The main objection to this system is that it is tied up with the primary feeder making it necessary to provide two triple-pole network switches for protection. The translator is independent of the primary voltage. The fourth transformer is greater in kv-a. rating than the translator and very much more expensive in case of 13,200-volt or 11,000-volt primaries. The fourth transformer should be located in the same manhole with the other three while the translator may be located where convenient. Part of the lighting load is supplied from open-delta connections.

DISTRIBUTION TO SUPPLY INCREASING LOAD DENSITIES¹

(CRAWFORD)

SEATTLE, WASHINGTON, SEPTEMBER 18, 1925

C. A. Heinze: Mr. Crawford assumes as an ultimate load density in his residential areas approximately 6250 kw. per sq. mi. He has used some figures that apply apparently to Seattle, which possibly may not apply to other cities. The number of houses per block, and the number of blocks per square mile—those are local conditions. The ratio of apartment houses and flats to single residences determines the load density for residential consumers. Studies of the conditions as they exist in Los Angeles show that it is reasonable to expect future load densities of 2500 kw. per sq. mi. for single residences, 6000 kw. per sq. mi. for flats and 19,000 kw. per sq. mi. for apartment houses. The actual load density in any square mile will be determined by the ratio between the different classes.

A few years ago the Commonwealth Edison Company of Chicago had already experienced in the Loop district a load density of 75,000 kw. per sq. mi., and was preparing for a future load density in this same area of 200,000 kw. per sq. mi.

The great majority of papers presented at this convention have to do with Y-connecting the present 2300-volt systems in order to get a higher feeder voltage. Personally I think that we are wrong in stopping at this voltage. We should go higher. While some authors have intimated in their papers that if they were to start over they would probably adopt higher voltages, it seems to me the sooner we adopt the highest practical voltage considering future load densities, the easier our future distribution problems are going to be. Certainly we do not want our systems growing larger and larger with the lower voltages, and then reach a time when we have to change to a higher voltage. Why not do the changing at the time when the system is small?

One thought that seemed to run through all the papers is that

each of the authors in turn said he has a system plan. Certainly that is a big step in the right direction. The trouble in the old days was that we did not have a system plan.

Having a system plan, and realizing the enormous load densities of the future, we should build today that proportion of the ultimate plan which will give us maximum economies.

It should be realized that when a 2300-volt delta-connected system is Y-connected, we have then reached the ultimate change that can be made in that particular system. On the other hand, by adopting a higher delta voltage for the time being, we may then ultimately, when our load densities become heavier, Y-connect that and have a voltage suitable for the densities of that time.

R. E. Cunningham: I want to ask Mr. Crawford what happens when their primary phase wire falls on the ground. We have had some very serious and expensive accidents; our primary wires have fallen on the ground and lain there some time before we discovered them, and in the meantime caused troubles. To us, taking care of that feature and fully protecting the public is more important than possibly saving a little copper by using a common ground return.

Our state law requires that where the higher voltage lines such as 11-kv. and 15-kv. circuits are paralleling telephone toll lines and become grounded, we must discontinue the service until the ground is cleared. From the hazard to the public it has become a practise with us, even though it does not parallel a toll line, to pull off the circuit until the ground is cleared. On a delta-connected system we have ground detectors which will show us when our circuit is on the ground, but with a Y-connected system such as the 4-kv. we are still hunting for the answer. If Mr. Crawford could, I wish he would give us what his experience has been and what is the accepted practise of taking care of cases where the primary wire falls on the ground.

We have only recently begun the use of automatic reclosing switches having installed two smaller substations with such equipment on the 4-kv. outgoing lines. We arbitrarily adopted the practise of reclosing the first time after 2-sec. interval, the second time, 15 sec. and the third time 30 sec. If the short circuit still remains on the circuit, the switch then locks out. I would like to know what Mr. Crawford's practise is in that regard.

L. R. Gamble: I note that Mr. Crawford has planned his system somewhat in the way the telephone engineers have been planning their systems for the past several years. It always seemed to me that it was a very good scheme. However, very few companies today are doing that thing, planning for the future.

Regarding the matter of diversity in load, on distribution circuits, I note Mr. Crawford has arrived at 1600 watts per residence. In accordance with tests now being made by the National Electric Light Association, on the Electric Range Committee of which I am the engineer in charge of collecting all the data and making the reports, we have arrived at a similar figure on our more or less superficial tests. The 1600 watts per residence is on the basis of about 150 residences, and takes into account the diversity so that in considering a distribution network such as Mr. Crawford has considered here, the figure is probably fairly close. However, in radio systems, that figure is going to be somewhat higher as the diversity factor will be smaller for a smaller number of residences.

I have been more or less a dyed-in-the-wool radial fan. I have not particularly liked the network system. As time goes on, however, it seems that due to improvements in apparatus, and varied schemes in working out different problems in practise, the network system for the higher-density loads is the coming system. No doubt it is in underground systems, but in overhead systems, there seems to me to have been some sort of question.

There is one question I wanted to ask Mr. Crawford. In laying out his system for the future, he figures that he will not have

¹ A. I. E. E. JOURNAL, October, 1925, p. 1063.

to rebuild or make any very material changes as the load grows. Considering this, does he use the butt-treated pole? Of course, I realize in asking this that not only does the butt of the pole decay in certain localities, but sometimes the top is also subject to rot.

There is one other point, and that is regarding trees. By the extension of the primaries to a considerable length and interlacing them as Mr. Crawford does, he is subjecting his line to tree interference which is very considerable. All distribution engineers and power men realize the disadvantage of trees, and, so long as we are going to have the "city-beautiful" idea, which luckily a great many cities have not yet adopted, our troubles will be ever increasing.

Digressing a little from Mr. Crawford's paper, there is one subject which should be of interest to distribution engineers and that is the painting of poles. An engineer is an artist to a certain extent; probably in the negative sense. The idea is that he should be able to put these poles in the streets and paint them such a color as to paint them out of the picture instead of into it. Of course, we don't all paint our poles. In some places it is required by ordinance. But I think we ought never to paint a pole a pea-green color.

F. H. Mayer: It has been found in designating electrical grounds for substations and power houses that it is more practical to ground non-current-carrying structures by means of a network, supplemented by one or two ground electrodes, than to ground the various structures by numerous electrodes.

A well designed grounding system must serve two general functions—

1. It must be designed so that its operator can come in contact with any non-current-carrying structure with safety at any time, and

2. It must permit of continuity of service.

The safety feature is accomplished by the metallic network, because a condition is approached similar to that of a station that is constructed on an immense metallic plate, thus tending to keep the potential gradient at the time of an electric failure to practically zero. The control of the potential gradient permits of the grounding of 220-kv. transformer neutrals and cases to the network with absolute safety to the operators.

The continuity of service is brought about by a more reliable relay performance and due to the fact that there is a low-resistance path from any part of the structure there is no tendency for high-voltage current to pass through the control board or other vital parts of the station.

So far as local failures are concerned the network suffices. However, if failures occur foreign to the station, the ground current returns to the station transformer neutrals through the earth and finally is picked up by the electrode that supplements the network, or if an overhead ground wire is used some of the current will return over it.

The overhead ground wire may not be essential where it is possible to extend the ground electrode into a stratum of permanent moisture and it is reasonably sure that the same stratum is again cut at the other stations. This is not always practical in rough areas, and in such cases the overhead ground wire serves a very good purpose, in that it furnishes a lower-resistance path than what is possible through the earth and ground electrode. Most of our power houses are located in mountainous country and such is our experience.

The point that can be gained, I think, is that on distribution systems feeding consumers in a hilly section, for the safety of human life, it is of prime importance, to furnish some reliable metallic return. This can be accomplished by grounding all transformer cases and neutrals to the water pipe or by conductors as outlined by Mr. Crawford.

It has been found that water pipes are not always reliable. Our experience showed in one instance that the pipe supplying the operator's cottage with gas was of lower resistance than the

water system. In this particular case, the water system was connected to the transformer cases through the transformer-cooling system. The gas main was connected to the water main through the automatic water heater and at the time of an electrical failure on the 60-kv. switching rack the ground current that flowed through the water heater was sufficient to destroy the heater coil.

It may be possible that this water system also had insulated sections. At any rate, it only goes to show that water-pipe grounds are not always reliable.

M. T. Crawford: I have been very much pleased at the evident appreciation of the policy of looking into the future as far as possible. I believe it is a fundamental responsibility for us, as engineers, to do this, inasmuch as the managements of our respective companies rely on us to a large extent in connection with making heavy investments which are not readily changed.

Mr. Gamble raised a question in regard to the use of networks as a general principle. That seems to me one of the fundamental necessities if we are to build for the future. Five years ago the Puget Sound Power and Light Company installed an a-c. low-voltage network in the Seattle underground district, which was the first one in the country to employ the principle of power-directional relay protection for interconnecting transformers on different feeders on the same system of mains. After these years of operation, we have felt that it is justified, and therefore, the network principle was considered in connection with future plans for the residential areas.

We had in mind building a distribution network something similar in principle to a fishnet, which might be suspended at a certain elevation above the floor. As various articles were thrown on this fishnet, tending to sag it down at certain points, supports in the form of wire or rope from the ceiling could be installed near those points, which would bring it back to level. They correspond to feeders from the substation, running out to tap the network where the load is heavy, bringing the voltage of the network up to the normal value.

Mr. Heinze referred to Dr. Ryan's prediction of one kilowatt per capita and interpreted that to result in approximately 19,000 kw. per sq. mi., and compared it with the 6250 kw. per sq. mi. load density mentioned in this paper as presently to be anticipated load.

I think that it should be borne in mind that the one kilowatt per capita or 19,000 kw. per sq. mi. includes downtown commercial loads, industrial power, and other loads in the city, whereas the figure of 6250 kw. in the paper refers solely to the residential-lighting and domestic load. In the same areas 13,000-volt loop feeders are run for large industrial power customers and as stated in the paper, their loads are not included in this load-density figure. Data are being collected now by the a-c. low-voltage network subcommittee of the National Electric Light Association, and I have in my hands for the report of that committee load-density figures from various sections of the country in the larger cities and the figures of 19,000 kw. per sq. mi. would be very low indeed unless it was an average of both light and heavy load densities, inasmuch as in the central part of cities like Chicago and New York the load densities aggregate 190,000 to 200,000 kw. per sq. mi. The area immediately surrounding this hotel in Seattle now has a load density of over 100,000 kw. per sq. mi.

Mr. Cunningham asked about our experience with one wire on the ground during trouble. We have had both experience with the delta system, ungrounded, and more recently with the grounded neutral in this respect. We find that grounded-neutral system is very much more apt to trip out the feeder at the station, in case of a feeder on the ground. In fact, with the delta system, we have had partial grounds persist three or four days at a time, which we could not locate, which were not apparently causing any dangerous condition, and sometimes cleared themselves up;

but on the grounded-neutral system, if we get a ground at all, it is very apt to build up into a serious one in a short time, causing the feeder to trip out.

Mr. Cunningham also asked about the re-closing features in our automatic substations on these grounded-neutral 4500-volt feeders. At the present time we have the feeder switches set to reclose after 30 sec. when they trip out from overload and after reclosing three times they lock out. Then, of course, we have to find the trouble on the feeder before it can be put back in service. In a considerable proportion of the cases, however, before the third trip-out occurs, the short circuit either burns itself entirely clear, or for some other reason is broken, so the feeder stays in after the first or second reclosure.

Mr. Gamble asked about butt-treatment and painting of poles. We have not gone into butt treatment to any considerable extent as yet, although a good deal of investigation has been under way, and we have had sample carloads of poles treated several ways, not only with butt-treatment, but with full pressure treatment of the pole as a whole. On Puget Sound, however, conditions are probably the least favorable for economy in connection with the butt-treatment or other treatment of poles, inasmuch as the cedar grows locally and is always obtainable at a lower price than in most other parts of the country; furthermore, we have found that where cedar is reset as a pole in the same soil in which its native growth occurred, it lasts very much longer, and we have thousands of poles which are now 20 years old which were not treated at all originally, and which still have sufficient strength, although the average life will probably not exceed 18 years. Where poles will live 18 to 20 years without treatment, it is a grave question as to whether a large increase in initial investment is to be incurred from which no return will be obtained until at least 18 to 20 years later. As other changes occur, such as the building up of the country, perhaps underground requirements may necessitate the removal of a considerable number of the poles within 20 years.

We paint all poles set in the city a very dark green except the 6 ft. next to the ground, which is painted black. With this arrangement, poles set up through shade trees are very unobtrusive. The tree-interference problem with us is a serious one, as with Mr. Gamble and others, but we have an arrangement whereby once each year we organize a tree-trimming crew which works under the supervision of a representative from the city park board, and the city ordinance permits us to trim trees in parking strips; in fact, requires that it be done by the power company. The trimming work done in this way tends to keep the trees at least out of the primaries.

Mr. Mayer pointed out in the discussion an instance of a water-pipe ground which did not prove reliable. But I think that this is something which it is quite important to emphasize here, and it is not so much the reliability of any one water-pipe ground as it is the reliability in the aggregate of a large number of such grounds. The grounding is done on the 4500-volt grounded-neutral system, primarily as a safety precaution, not as a means of returning neutral current, and in any construction or design for the purposes of safety precaution, a multiplicity of small precautions are more reliable than one single large one. There was a case not long ago in Victoria of a fatal injury where the court, after hearing considerable expert testimony, was of the opinion that the injury was primarily the result of water-pipe grounds not having been made. The local light and power company had made as good grounds as they could, but the soil being very rocky these grounds were entirely inadequate. The local authorities would not permit them to make water-pipe grounds on individual services. After collecting considerable data and testimony, it was the opinion of the court that such grounds should have been provided as a precaution for the safety of the public.

SPANS HAVING SUPPORTS AT UNEQUAL ELEVATION¹

(SMITH)

SEATTLE, WASHINGTON, SEPTEMBER 15, 1925

F. K. Kirsten: I wish to congratulate Mr. Smith on his courage in using the catenary equations for the analysis of spans supported from points at unequal elevations, and I would also like to make some supplementary comments.

We know that the more closely we wish to describe a physical phenomenon by means of mathematical formulas, the more involved and unwieldy these formulas become. As an illustration, we might use the simple equation of a circle to express the location in space of a suspended cable, and describe actual conditions with sufficient accuracy although the assumptions involve considerable error. By a closer analysis of the forces in action we find, however, that the parabola describes actual conditions better than the circle and hence the application of the parabola yields greater accuracy, although the mathematical operations are more involved. But still a considerable error is made in the assumption that the weight per unit projected length of the span is uniform along the span. The demand for greater accuracy forces us to base our mathematical formulas upon the assumption that the weight per unit length of span, measured along the span, is constant. This assumption leads us to the catenary, a rather involved mathematical stratagem. This mathematical form is rather difficult to handle, especially if temperature changes accompanied by stress changes must be covered by its use. And still the catenary equation does not describe conditions with absolute accuracy. Since the tension changes along the span, the weight per unit length of cable cannot be uniform. But an attempt to involve this actual condition in the modification of the catenary form would lead to unmanageable expressions.

It will be apparent from a perusal of Mr. Smith's work that the chief difficulty in applying the catenary equation to spans of unequal elevations resides in the introduction of the slope of the suspension points into equations which are already complex enough for ready manipulation. Especially do we feel a naturally increasing reluctance to use the catenary analysis if changes of wind and ice loading together with temperature variations over a considerable range must be accounted for by proper mathematical operations with the catenary form. It is, therefore, in my opinion, a step in the right direction to adhere to the simple catenary forms which Mr. Smith ingeniously introduces into his second method instead of using the first method supported by equations 12, 14 and 15.

We now have, thanks to Mr. Smith's work, a simple catenary analysis of spans of unequal elevation, and there cannot be any excuse for the use of parabolic forms in the future. It must be remembered, that the catenary method is independent of fixed mass and space units, hence all span conditions which may occur in practise may be expressed by a series of curves from which by interpolation any span, at any slope, for any size or material of cable or any temperature range, may be read at once. This set of curves is as readily applied, without modifications by constants, in continental Europe where the metric system is used as in the lands of inches, pounds and quarts.

Another important finding in Mr. Smith's paper I believe is the discovery that the point of maximum deviation from a straight line connecting any two points of unequal elevation of the cable in suspension occurs midway between the points of support in the middle of the span. This naturally facilitates a check upon the strain of a cable from points of unequal elevation by dropping targets from point B_2 to point A , of predetermined length, the line of vision touching the point of maximum deflection from the line joining these two points.

R. J. C. Wood: Our engineering department has done

1. A. I. E. E. JOURNAL, December, 1925, p. 1352.